

Absolutely. At this point the project has evolved considerably beyond the original NVIDIA-focused harness, so the checkpoint should reflect both:

- 1. The engineering progression
- 2. The architectural purpose of each stage
- 3. The final state after J1.5

Agent Harness Research System (v1.1)

Complete Project Readout Through J1.5

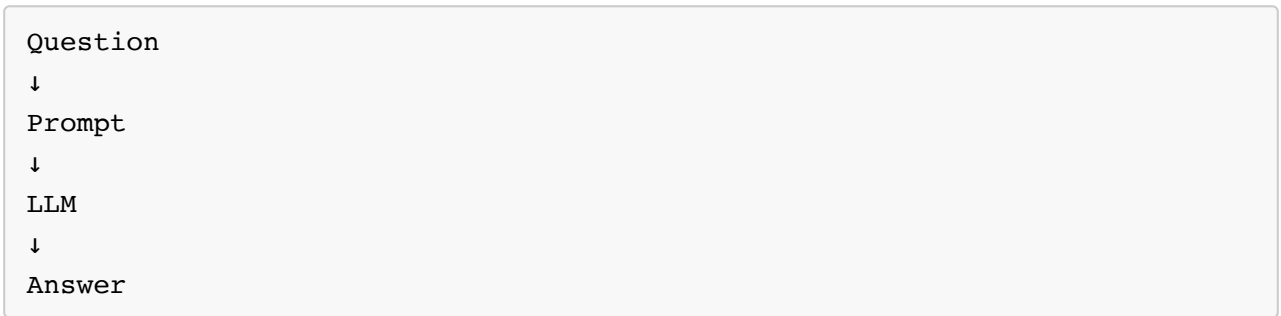
Executive Summary

The project began with a simple question:

How do we move from prompting an LLM to building a reliable research system?

The answer was to implement a **Harness Engineering** architecture.

Rather than relying on:



the harness evolved into:

```
Question
↓
Domain Profile
↓
Retrieval
↓
Evidence
↓
Contradictions
↓
Research Gaps
↓
Coverage Analysis
↓
Evaluation
↓
Trace
↓
Memo
```

The final system is capable of:

- Multi-document research
- Evidence extraction
- Evidence ranking
- Source quality assessment
- Contradiction detection
- Research gap detection
- Coverage analysis
- Domain portability
- Regression testing
- Full traceability

Core Design Philosophy

The harness treats the LLM as:

```
One component
```

rather than:

The entire system

The harness itself provides:

- workflow
- governance
- observability
- quality control
- repeatability



Current Architecture

```
Question
↓
Domain Profile
↓
Topic Detection
↓
Document Loading
↓
Chunking
↓
Retrieval
↓
Evidence Extraction
↓
Source Weighting
↓
Contradiction Detection
↓
Research Gap Detection
↓
Coverage Analysis
↓
Evidence Ranking
↓
Memo Synthesis
↓
Evaluation
↓
Trace Generation
↓
Regression Tests
```

Session 1

Initial Research Harness

Objective

Build the smallest useful research workflow.

Architecture

Documents

↓

Evidence

↓

Memo

Output Sections

- Executive Summary
- Confirmed Facts
- Inferences
- Power Implications
- Cooling Implications
- Open Questions

Result

Created the first end-to-end research pipeline.

Step A

Observability

Objective

Make system behavior visible.

Added

Debug mode:

```
--debug
```

Source reporting:

```
--show-sources
```

Trace generation:

```
memo.trace.json
```

Result

The system became inspectable.

Every major operation became visible.

////////////////////////////////////

Step B

Claude Integration

Objective

Replace mock outputs with real model reasoning.

Added

Claude Sonnet integration.

Operations:

```
Research Planning  
Evidence Extraction  
Memo Synthesis
```

Result

Research quality became model-driven rather than template-driven.

////////////////////////////////////

Step C

Source Grounding

Objective

Ensure all conclusions are traceable.

Added

Evidence IDs:

E001
E002
E003
...

Source attribution.

Chunk attribution.

Result

Every claim became traceable to source material.

////////////////////////////////////

Step D

Question-Aware Evaluation

Objective

Evaluate outputs against the actual research question.

Previous

Same evaluation for every question

New

Question-specific expectations.

Example:

```
Cooling question
→ Cooling coverage required

Networking question
→ Networking coverage required
```

Result

Evaluation became context-aware.

////////////////////////////////////

Step E

Regression Testing

Objective

Prevent quality regressions.

Added

Evaluation corpus:

```
eval_001
eval_002
eval_003
...
```

Evaluation reports.

Result

Changes became measurable.

////////////////////////////////////

F1

Evaluation Preservation

Objective

Keep historical evaluations.

Added

Persistent evaluation outputs.

Result

Performance could be compared over time.

F2

Evidence Ranking

Objective

Not all evidence should be treated equally.

Added

Evidence scoring:

Relevance
Source Quality
Specificity

Combined:

Overall Score

Result

The memo became driven by stronger evidence.

F3

Chunking

Objective

Scale beyond a few PDFs.

Previous

```
Entire document
↓
LLM
```

New

```
Document
↓
Chunks
↓
Evidence
```

Result

Large corpora became practical.

F3.1

Chunk Diagnostics

Objective

Explain chunk-level behavior.

Added

Per chunk:

- relevance score
- evidence count
- extraction decision
- rejection reason

Result

Chunk behavior became observable.

////////////////////////////////////

G1

Question-Aware Retrieval

Objective

Process only relevant chunks.

Previous

```
All chunks
↓
Evidence extraction
```

New

```
Question
↓
Chunk Ranking
↓
Top Chunks
↓
Evidence Extraction
```

Result

Better scalability and lower cost.

////////////////////////////////////

G2

Contradiction Detection

Objective

Identify conflicting evidence.

Example

```
120 kW
vs
180 kW
```

Added

Contradiction records:

- severity
- confidence
- explanation

Result

Conflicting claims became visible.

////////////////////////////////////

G3

Research Gap Detection

Objective

Identify missing information.

Example

```
No UPS requirements found
No heat rejection figures found
```

Added

Gap records:

- topic
- priority
- rationale

Result

The system can recommend future research.

////////////////////////////////////

G3.1

Synthesis Scaling

Objective

Prevent memo generation failures.

Problem

Too much evidence caused:

```
max_tokens exceeded
```

Added

- token accounting
- evidence budgeting
- synthesis diagnostics

Result

Large evidence sets synthesize successfully.

////////////////////////////////////

H1

Source Quality Weighting

Objective

Teach the system that sources differ in credibility.

Example

```
NVIDIA Technical Spec
>
Vendor Marketing
>
Synthetic Test File
```

Added

Source quality scores.

Used By

- Retrieval

- Evidence Ranking
- Contradiction Analysis

Result

More trustworthy evidence drives conclusions.



H2

Coverage Matrix

Objective

Measure topic coverage.

Distinction

Research Gaps:

What is missing?

Coverage Matrix:

How much evidence exists?

Example

Power	Strong
Cooling	Moderate
Networking	Weak

Result

Coverage and gaps became separate concepts.



J1

Domain Profiles

Objective

Separate:

```
Research Engine
```

from:

```
Domain Knowledge
```

Previous

AI infrastructure concepts were embedded throughout the system.

New

Profiles:

```
profiles/  
  ai_data_centers.yaml  
  smr.yaml
```

Result

The same harness can support multiple domains.

//

J1.1

Profile-Based Topic Detection

Objective

Detect topics using profile definitions.

AI Data Centers

Power
Cooling
Networking
Operations

SMRs

Licensing
Construction
Economics
Fuel Cycle
Grid Integration

Result

Topic detection became domain-specific.

////////////////////////////////////

J1.2

Profile-Based Research Gaps

Objective

Use profile-defined gaps.

Example

AI Infrastructure

UPS
PDU
Heat Rejection

SMRs

LCOE
Overnight Cost
Construction Permit
Fuel Supply

Result

Gap detection became portable.

////////////////////////////////////

J1.3

Profile-Based Source Classification

Objective

Allow source quality to vary by domain.

SMR Example

Score 5

DOE
IAEA
NRC
NEA
INL

Score 4

NuScale
TerraPower
GE Hitachi
World Nuclear Association

Result

Source weighting became domain-aware.

J1.4

Contradiction Normalization

Objective

Prevent false contradictions.

Problem

The system incorrectly flagged:

```
300 GW target  
vs  
13 GW/year licensing throughput
```

Solution

Add:

```
Entity Matching  
Metric Matching  
Unit Normalization  
Scope Awareness
```

Result

Only genuine contradictions remain.

Synthetic Test Suite

Correctly detected:

24–36 months
vs
8–12 years

Correctly ignored:

HALEU OECD
vs
HALEU Russia/China

300 GW
vs
13 GW/year

Future modular benefits
vs
FOAK history

J1.5

Profile-Aware Contradiction Taxonomy

Objective

Remove domain-specific contradiction labels.

Previous

SMR contradictions could still appear as:

rack power

New

SMR contradictions use:

```
construction
licensing
economics
fuel_cycle
grid_integration
```

Example

```
{
  "topic": "construction",
  "topic_source": "profile:smr"
}
```

Result

Contradiction categorization became domain-aware.

////////////////////////////////////

Final Capability Status

Capability	Status
Document Loading	✓
Evidence Extraction	✓
Source Grounding	✓
Evaluation	✓
Trace Generation	✓
Regression Testing	✓
Evidence Ranking	✓
Chunking	✓
Retrieval	✓
Contradiction Detection	✓
Research Gaps	✓
Coverage Matrix	✓
Source Weighting	✓
Domain Profiles	✓
Domain-Aware Contradictions	✓



What We Proved

The most important outcome is not a specific feature.

It is this:

The harness successfully ran on:

AI Infrastructure

NVIDIA Rubin
Power
Cooling
Racks
Networking

and then on:

Small Modular Reactors

Licensing
Construction
Economics
Fuel Cycle
Grid Integration

without changing:

- chunking
- retrieval
- evidence extraction
- synthesis
- evaluation
- tracing

Only the profile changed.

That is the hallmark of a successful Harness Engineering implementation.

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Current Decision Point

With J1.5 complete, the next major directions are:

K1 — Multi-Domain Evaluation Suite

Build formal regression suites for:

- AI Infrastructure
- SMRs
- Grid Infrastructure

- Semiconductors

to validate portability.

I1 — Document Acquisition

Add:

```
Question
↓
Search
↓
Download
↓
Sources
↓
Harness
```

to automatically discover and ingest documents.

My recommendation would be **K1 first**, because the harness architecture is now mature enough that broader validation will teach you more than adding another capability.